

CHALLENGER BRIEFING • WEEK 4 PROJECT

Clone *Synthesis* Tutor.

Five days. One lesson. Build a conversational AI tutor that makes a kid *feel* a math concept click.

PROJECT	WK04-CST
CONCEPT	FRACTION EQUIVALENCE
DEMO	FRIDAY • NOON

THE MISSION

Build a single math lesson that feels like *exploration*, not homework.

THE MEDIUM

A chat tutor + an interactive manipulative, side by side.

THE BAR

A 9-year-old runs it on an iPad and gets it.

THE CLOCK

Five days. Demo by Friday at noon.

BRAINSTORMING • NUDGES, NOT A MAP

Don't start with code. *Start with the moment.*

01

Play it first.

Spend an hour as the kid. Open Synthesis. Notice the exact moments you lean in.

02

Pick the aha.

What's the smallest revelation worth a whole lesson? Build outward from that one beat.

03

Steal the gesture.

What's your "smash"? Pick one direct-manipulation move and make it feel inevitable.

04

Script the miss.

What does the tutor say when they get it wrong? The recovery is where the warmth lives.

05

Design the win.

What does "I got it" look, sound, and feel like? End on a high or the lesson didn't land.

06

Paper before pixels.

Sketch the full lesson on paper before you write a single line of code. It's faster. Trust it.

THE CHALLENGE

Replicate *the magic*, on a one-week clock.

You are not building a curriculum. You are not building a platform. You're shipping a single, self-contained lesson that proves the model — a chat tutor that talks like a person, and a manipulative a kid wants to poke.

The interaction quality is the work. The scope is your friend.

FROM THE PRD

"It creates a learning experience that feels like *exploration rather than homework*, successfully engaging children and building deep conceptual understanding."

PRD • PROBLEM STATEMENT • V1

WHAT YOU'RE BUILDING

Three pieces. One lesson.



01 • Conversational Tutor

A chat that teaches.

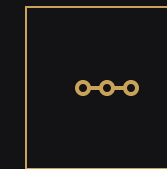
Scripted dialogue — no LLM required. Warm tone, simple branching on right/wrong answers. The tutor is the teacher.



02 • Interactive Manipulative

A toy that teaches.

A fraction box the student can poke, combine, split, and "smash." Direct manipulation is the lesson — make the math feel physical.



03 • Lesson Flow

A path with a finish.

Explore → instruct → check. The student plays first, then answers, then proves they got it. The lesson ends when they win.

THE CONCEPT

Fraction *equivalence*.

One half is the same as two quarters. Obvious to you. A small revelation to a third-grader.

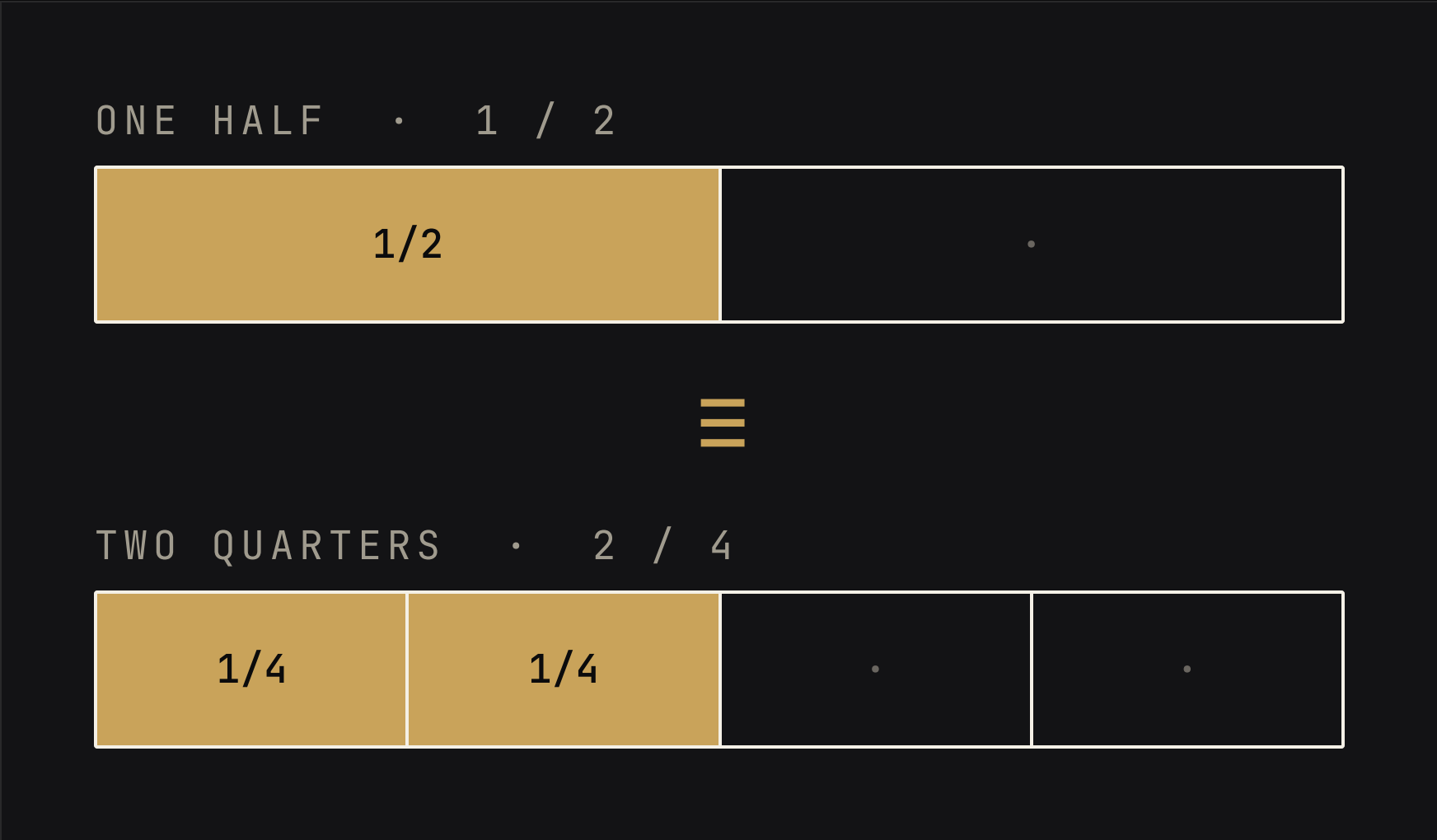
Make them see it, slide it, and smash it together — until the equals sign is the only place it could land.

GRADE 3

FRACTIONS

VISUAL + TACTILE

IPAD-FIRST



YOUR WEEK

Five days. One demo.



SHIP IT

What lands by *Friday noon* — and how it's judged.

DELIVERABLES

What you turn in.

01 A deployed web app.

Single lesson on fraction equivalence, live on a public URL, working in a modern browser.

02 An iPad roadmap.

A short doc + sketches: how the lesson works on iPad. Touch targets, gestures, what changes from desktop.

03 A 3–5 minute demo video.

Show the conversational flow and the manipulative in action.

04 A README.

How to run it, plus a short note on your technical approach.

HOW WE JUDGE

What we're looking for.

→ Interaction quality.

Does the tutor feel warm? Does the manipulative feel alive? This is the work.

→ Lesson coherence.

Explore → instruct → check. Branching on wrong answers. A clean finish.

→ Demo-readiness.

No setup hell. No "imagine if." The web app runs, and the iPad story is convincing.

FLOOR'S YOURS

Ask *anything.*

Then go build something a kid would want to play with on a Saturday.

CLOCK STARTS MONDAY 09:00 • DEMO FRIDAY NOON →